

KRISHNA DEEP YERRAMALLU

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Education

Stevens Institute Of Technology

Masters of Science in Computer Science

New Jersey, USA

Graduation: December 2025

Jawaharlal Nehru Technological University

Bachelor of Technology in Computer Science and Engineering

Hyderabad, India

Graduation: August 2022

Experience

Software / Data Engineer - Data & Platform Team

June 2025 - Present

Nerxwave

Remote, USA

- Designed and implemented a **self-service data platform** using **standardized ingestion templates**, medallion architecture, using **Databricks and PySpark**, allowing analytics and product teams to **consume curated datasets** without direct engineering involvement.
- Established **data SLAs** using (**CDC**), **schema versioning**, and **automated data quality checks** using **SQL, Python and dbt**, with pipeline monitoring integrated into **CI/CD workflows** to reduce downstream data failures.
- Modeled analytics-ready **fact and dimension tables** using **star-schema and incremental loading**, and delivered **Tableau and Power BI** dashboards consumed by product teams and executives for recurring business reviews and operational tracking.

Analyst - Data Warehousing

August 2022 – August 2024

Silicon Labs

Hyderabad, India

- Architected scalable data pipelines in **Azure Synapse, Databricks** using **Python and Go** for **RPA, ETL** and streamline **Apache Airflow** data orchestration, **reducing human intervention by more than 75%**.
- Optimized data processing and movement across layers in a **Medallion Architecture** using **advanced SQL patterns** like **Change Data Capture (CDC)**, **reducing Azure egress costs by 40%** from 2023 to 2024.
- Developed and **Dockerized** a system to **synchronize cloud and on-premise databases** using **CDC** and **row-level hashing**, enabling efficient incremental updates and **minimizing egress costs** through selective data transfer.
- Collaborated with data scientists, analysts, over 8 business units to **design data integration solutions**, developing **domain-specific data marts** to support diverse **analytical and machine learning use cases**.
- Implemented **automated data quality and governance frameworks**, integrating validation checks, schema integrity rules, and lineage tracking to **ensure 99.9% data accuracy and compliance** with enterprise access policies.
- Developed and automated **Power BI and Tableau dashboards** to monitor **cloud migration progress, post-migration system performance**, and **data quality metrics** (nulls, schema drift, duplicates), enabling leadership to track KPIs in real-time, **reduce ad-hoc queries**, and **accelerate incident resolution**.
- Migrated 200+ legacy applications, data sources, and reports to the cloud using **Python, SQL, Kafka, Spark**, and **Terraform**, as part of a **cloud migration project**, resulting in a **50% reduction in report generation time**.
- Implemented **CI/CD pipelines** using **Jenkins, Redgate, Azure Pipelines, Git** to **automate SQL database schema changes**, enabling smooth and **version-controlled deployments** across environments.

Research Assistant - Machine Learning and Deep Learning

January 2020 - January 2022

Keshav Memorial Institute Of Technology

Hyderabad, India

- Developed an **AI-powered pathology platform** deployed in diagnostic centers, assisting pathologists in breast cancer detection and grading, improving **diagnostic accuracy by 97%**.
- Built a **web-based WSI analysis tool** adopted by clinical teams, reducing diagnostic **turnaround time by 30%** while maintaining **94% accuracy**.
- Technologies used:** Tensorflow, Pytorch, Hugging Face, cuDNN, Nvidia DGX, scikit-learn, Fast API.

Technical Skills

Languages: Java, Python, C/C++, Go, SQL, Bash

AWS Services: EKS, EC2, RDS, IAM, Lambda, S3, ElasticCache, MSK

Azure Services: Data Factory, Virtual Machine, SQL Database, Functions, Blob Storage

DevOps: Docker, Kubernetes, Git, Jenkins, Terraform, CI/CD, Linux

Data Processing: Kafka, Spark, Snowflake, MySQL, PostgreSQL, Apache Airflow, Excel

Machine Learning & AI: Scikit-learn, TensorFlow, PyTorch, Hugging Face, LangChain, RAG, Transformers

Visualization: Tableau, PowerBI

Publications & Achievements

- DCS_PathIMS: AI powered Digital Pathology Diagnostics Platform for Breast Cancer Histology Imaging Biomarker Discovery for Precision Oncology [link](#).
- Awarded the “**Most Innovative Hack**” at Stevens QuackHacks 2025.